
Modular Programming – Practical Work

Week 5

Classes and Objects, a deeper look 2

Notations: W1Q1 = Week 1, Questions of type 1

Question types:

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|---|----------------------|
| • Type 1: theory, concepts | <i>knowledge</i> |
| • Type 2: understanding typically based on code reading | <i>understanding</i> |
| • Type 3: producing yourself code | <i>know-how</i> |

A moodle deposit space is open every week for you to upload your solutions. Solutions of the exercises are provided one week later.

Important points:

- 602_F and 602_D classes should upload solutions **before Monday 24h00 the following week**
- **2 or 3 practical work will be corrected** and will count for one quarter of the intermediate test note

W5Q1 – 1

1. What is the difference between inheritance and composition?

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2. What is the difference between `final` and `finalize`?

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3. *Casting*: What is an implicit *cast*? What is an explicit *cast*? In which case an explicit *cast* will generate an error

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4. Explain the operation of the access modifier protected ?

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5. Why and when should we use the instanceof operator? Can we apply the instanceof operator to primitive types ?

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W5Q2 – 1 IMPLICIT CAST

Have a look at the following code :

```
int number_1 = 7;
int number_2 = 3;
float number_3 = 7;
float number_4 = 3;
int result_1 = 0;
float result_2 = 0;

result_1 = number_1 / number_2;
result_2 = number_1 / number_2;
result_1 = number_3 / number_2;
result_2 = number_3 / number_2;
result_1 = number_1 / number_4;
result_2 = number_1 / number_4;
result_2 = number_3 / number_4;
```

- Which of these statements do not compile? Why? What can be done to resolve the problem (to compile the program)?
- What are the results of the above calculations? Which calculations do not give the result 2.3333333? Why?
- If you substitute the digits 7 and 3 for the variables in statement two, which three different ways are there to change this statement so that it results in the correct value?

W5Q2 – 2 By value / by reference

Have a look at the following code :

```
class MyInteger
{
    private int value;
    public MyInteger( int value ) { this.value = value; }
    public void setValue( int n ) { this.value=n; }
    public int getValue() { return this.value; }
    public String toString() { return "" + value; }
}

public class Test {

    public static void main(String[] args) {
```

```

// Pt.1. passing by value
int a = 2;
changeSimpleType( a );
System.out.println( "a = " + a );

// Pt.2. passing by reference
Integer b = new Integer( 2 );
changeIntegerType( b );
System.out.println( "b = " + b );

// Pt.3. passing by reference (bis)
MyInteger c = new MyInteger(2);
changeMyIntegerType( c );
System.out.println( "c = " + c );
}

public static void changeSimpleType( int a )
{
    a = 10;
}

public static void changeIntegerType( Integer a )
{
    a = 10;
}

public static void changeMyIntegerType( MyInteger a )
{
    a.setValue( 10 );
}
}

```

Question 1: What is the value of variable `a` after a call to the method `changeSimpleType()` ?

Question 2: What is the value of variable `b` after a call to the method `changeIntegerType()` ?

Question 3: What is the value of variable `a` after a call to the method `changeMyIntegerType()` ?

Question 4: When you run this code, what do you remark? Why does the value of `b` remain unchanged even if it is passed as a reference? Check the methods of the `MyInteger` and the `Integer` classes to help you find the answer.

W5Q2 – 3 This

What is the output of the program `TestCircle` below? (answer without copying the code in Eclipse).

Note: Be careful using of `this` because here it is automatically transformed into a call to `toString()` which is redefining itself in the superclass and subclass.

```
/**
 * Point class represents a basic x-y coordinate pair
 *
 * @author jean.hennebert@hevs.ch
 */
public class Point {

    private int x; //x coordinate
    private int y; //y coordinate

    //no argument constructors
    public Point() {
        x = 0;
        y = 0;
        System.out.println("Point constructor (no argument): " + this);
    }

    public Point(int xValue, int yValue) {
        x = xValue;
        y = yValue;
        System.out.println("Point constructor: " + this);
    }

    public int getX() {
        return x;
    }

    public void setX(int x) {
        this.x = x;
    }

    public int getY() {
        return y;
    }

    public void setY(int y) {
        this.y = y;
    }

    public String toString() {
        return "[" + getX() + "," + getY() + "]";
    }
}
```

```

/**
 * Circle class declaration
 * @author jean.hennebert@hevs.ch
 *
 */
public class Circle extends Point {

    private double radius;

    public Circle() {
        setRadius(0.0);
        System.out.println("Circle constructor (no arguments): " + this);
    }

    public Circle(int xValue, int yValue, double r) {
        super(xValue, yValue);
        setRadius(r);
        System.out.println("Circle constructor: " + this);
    }

    public double getRadius() {
        return radius;
    }

    public void setRadius(double r) {
        if(r < 0.0) radius = 0;
        else radius = r;
    }

    public String toString() {
        return "center=" + super.toString() + " radius=" + getRadius();
    }
}

public class TestCircle {

    /**
     * @param args
     */
    public static void main(String[] args) {

        Point p = new Point(5,10);
        Circle c2 = new Circle(p.getX(),p.getY(),2.0);

        Circle c1 = new Circle();

    }
}

```

W5Q3 – 1 Shipping

We wish create a freight costs java program. Goods are instances of the Merchandise class :

Figure 1: The source java class Merchandise

```
public class Merchandise {
    private int weight;
    private int volume;

    public Merchandise (int weight, int volume) {
        this.weight = weight;
        this.volume = volume;
    }

    public int weight () {/ / return the weight in kg
        return weight;
    }

    public int volume () {/ / return volume in dm3
        return volume;
    }
}
```

The merchandises are conveyed as cargo. Only 16 different merchandises could be added to a single cargo. The only public cargo's methods are:

add that adds a merchandise in the shipment if it is still possible.

cost that returns CHF in cents, the total cost of cargo.

A shipment is also characterized by the distance over which it is carried. This distance is an argument of the cargo constructor; it is expressed as a whole number of kilometers.

Be careful not to exceed cargo's cost limit. The cost must not exceed limit value described in those "cost formulas":

Type	Cost formula	Limit value
By river	distance * weight	300000
By road	4 * distance * weight	38000
By air	10 * distance * volume	80000

Question: Draw the class diagram and write the program.